

MODEL 2.8.4

Why This Probability?

Metrics & Interpretation Guide

Purpose

A practical reference for reading the new probability-explanation and Research Confirmation fields in the MLB MODEL. It explains what each number means, how it relates to the final probability, and what deserves attention when the model and real-world context disagree.

Version scope: MODEL 2.8.4 Research Workspace. This guide describes the interpretation layer around the current K and Moneyline models. The Research Confirmation Engine (RCE) is observational in this version: it audits the prediction but does not re-weight the model probability.

Core principle: model probability, research confirmation, and market price answer different questions. A high RCE confidence score is not the same thing as a high probability that a bet wins.

1. How to Think About the Screen

The new explanation view has three jobs: show the mathematical drivers of the prediction, show the real-world context around those drivers, and make contradictions visible before a wager is treated as actionable.

MODEL MATH -> predicts
RESEARCH CONFIRMATION -> audits
MARKET -> prices the outcome

The three layers should agree often, but they do not need to agree.

The three numbers people most often confuse

Metric	What it means	How to use it
Model probability	Estimated chance of the selected outcome according to the model.	Use for fair price, edge and EV.
Market probability	Probability implied by the available betting price after market processing/devig where applicable.	Compare with the model to measure edge.
RCE confidence	Strength/completeness of the research evidence behind the RCE verdict.	Do NOT interpret as win probability.

Important

A 61% model probability with RCE confidence 90/100 means: the model estimates 61% and the independent audit strongly supports that conclusion. It does NOT mean the bet has a 90% chance to win.

2. Strikeout Props: The Core Calculation

The K engine is easiest to understand as a chain. Workload determines how many batters a pitcher is expected to face. The matchup determines the strikeout probability per batter. Those two produce Structural K, which is blended with recent and season strikeout information to create xK. A probability distribution around xK then produces the final Over/Under probability.

Expected workload -> Expected BF
Pitcher skill + opponent profile -> Matchup K%
Expected BF x Matchup K% -> Structural K
Structural K + Recent K + Season K -> xK
xK + uncertainty/dispersion -> P(Over) / P(Under)

xK - Expected Strikeouts

xK is the model's expected number of strikeouts for this pitcher in this specific game. It is the central K projection. It is not the sportsbook line and it is not simply the pitcher's season average.

Example
Market line: 4.5 K
Model xK: 3.37
Gap: -1.13 K

The probability engine asks how often a pitcher with this expected distribution finishes below 4.5.

What to inspect first

When a K probability looks unusually high or low, first compare xK with the market line, then inspect Expected IP/BF and Structural K. Those usually explain the projection faster than recent game logs.

Structural K

Structural K is the baseball-based strikeout expectation generated from today's projected opportunity and matchup. Conceptually it is Expected Batters Faced multiplied by the model's Matchup K rate.

```
Expected BF = 21.4
Matchup K% = 15.0%
Structural K ~= 21.4 x 0.150 = 3.21
```

Structural K is intentionally the largest piece of the xK blend because it describes today's matchup rather than relying mainly on noisy historical box-score totals.

3. Strikeout Props: Workload and Opportunity

Expected BF - Expected Batters Faced

Expected BF estimates how many hitters the pitcher is likely to face. A pitcher cannot strike out batters he never reaches, so BF is one of the most important opportunity variables in the K model.

The workload process uses structured pitcher markets and game context, including Pitcher Outs, Earned Runs, Hits Allowed and Walks where available. A leash adjustment can reduce expected outs when the run/hit/walk environment implies a greater chance of an early hook.

IP - Expected Innings Pitched

IP is the expected innings corresponding to the workload estimate. In simple terms, expected innings are expected outs divided by three. Low expected IP can suppress xK even for a high-strikeout pitcher.

```
Expected outs: 16.2
Expected IP: 5.4
```

Workload Status

Metric	What it means	How to use it
NORMAL	Expected workload is broadly consistent with a normal starter assignment.	Neutral/supportive for Overs; does not automatically oppose Unders.
LIMITED	Expected outs are materially below a normal full starter workload.	Important warning for Overs; often supports Unders.
OPENER_LIKE	Expected outs are very low and resemble an opener/bulk uncertainty profile.	Strong reason to investigate before trusting a K Over.

Current RCE thresholds are approximately: NORMAL above 14.5 expected outs; LIMITED from 8 to 14.5; OPENER_LIKE at 7.5 or below. These are research labels, not new probability weights.

4. Strikeout Props: Skill and Matchup

Pitcher K%

Pitcher K% represents the pitcher's underlying strikeout skill. The model does not fully trust small samples: current performance can be shrunk toward prior-season performance or a neutral baseline when the available sample is limited.

Opponent K%

Opponent K% attempts to describe the actual hitters expected to face the pitcher, not merely a generic full-season team rate. The projected/official batting order receives most of the weight, with a broader team tendency retained as a stabilizer.

In the current construction, lineup-specific K tendency receives roughly 72% of the opponent component and broader team tendency about 28% when both are available.

Matchup K%

Matchup K% is the estimated probability that an individual batter faced in this matchup becomes a strikeout. It combines pitcher and opponent tendencies and can also incorporate supporting swing-miss/contact information such as Whiff%, swinging-strike characteristics and Contact%.

Pitcher K% 27.0%
Opponent K% 24.0%
Matchup K% 28.6%

Interpretation: the interaction is especially favorable for strikeouts.

Opponent K Profile: Favorable / Neutral / Unfavorable

RCE translates the opponent strikeout environment into a direction relative to the selected bet. A high-strikeout opponent supports an Over but works against an Under; a low-strikeout opponent does the opposite.

Metric	What it means	How to use it
HIGH K	Approximately 24%+ matchup/opponent K environment.	Favorable to K Overs, unfavorable to K Unders.
NEUTRAL	Approximately 21% to 24%.	Neither side receives strong confirmation.
LOW K	Approximately 21% or lower.	Favorable to K Unders, unfavorable to K Overs.

5. Strikeout Props: Recent, Season and Blend

Recent K

Recent K is a recency-weighted view of recent strikeout production. Newer starts receive more weight than older starts, with the weight decaying by roughly 10% per start. It is not just a flat arithmetic average.

Season K

Season K is a longer-term strikeout production anchor from the structured pitcher history. It prevents a handful of recent starts from dominating the projection.

Blend Weights

When all three K components are available, the current raw weights are approximately 0.55 Structural, 0.15 Recent and 0.08 Season. After normalization, that is about 70.5% Structural, 19.2% Recent and 10.3% Season.

Structural K	3.20	~70.5%
Recent K	5.10	~19.2%
Season K	5.00	~10.3%

xK	blended result	

Why this matters

A pitcher can be averaging five strikeouts recently and still project much lower today if workload or matchup conditions materially reduce Structural K. The model is designed to prefer today's opportunity structure over box-score recency.

6. Research Confirmation for K Props

Last 10 and Last 5

These fields show how often the specific side and line would have won over the pitcher's recent history. Example: for Under 4.5, "8/10 Under" means eight of the previous ten tracked starts finished with four or fewer strikeouts.

These hit rates are research evidence. They are not simply added to the probability again, which would risk double-counting information already represented elsewhere.

Recent Form: HOT / AVERAGE / COLD

Recent Form describes whether the pitcher's current strikeout production is materially above, near or below his own baseline. The current RCE form signal is driven primarily by recent average versus season average, with a secondary comparison of the last three starts versus the previous three. Differences are normalized by the pitcher's usual variability.

Metric	What it means	How to use it
HOT ↑	Recent strikeout production is meaningfully above the pitcher's baseline.	Supports Overs more than Unders, but does not change xK in v2.8.4.
AVERAGE →	Recent production is broadly in line with baseline.	Neutral research signal.
COLD ↓	Recent production has materially weakened relative to baseline.	Supports Unders more than Overs; investigate the cause.

Recent K Avg

Recent K Avg is useful when compared directly with xK. A large difference is not automatically wrong, but it tells you the model is seeing something today that recent box scores do not.

Recent K Avg: 6.25
xK: 3.20

Question to investigate: what is creating a ~3 K reduction?
Possible causes: workload, opener role, lineup contact profile, handedness, stale input or role change.

7. RCE Signals, Confidence and Verdicts

Signal Icons

Metric	What it means	How to use it
✓ Support	Evidence supports the model-selected side.	Positive confirmation.
⚠ Conflict	Evidence points against the model-selected side.	Investigate before betting.
• Neutral	Evidence is inconclusive or not directionally meaningful.	No strong confirmation or rejection.

K research currently considers signals such as recent distribution, opponent K profile, Recent Form and workload. More feeds can be added later without changing the core prediction math.

RCE Confidence

RCE Confidence is a 0-100 audit-quality/support score. It reflects how strongly and completely the available research evidence supports the RCE verdict. It is NOT a probability that the bet wins.

Research Verdicts

Metric	What it means	How to use it
STRONGLY CONFIRMED	At least several meaningful signals support the model with no major contradiction.	Cleanest research alignment.
CONFIRMED	Multiple signals support the model with little contradictory evidence.	Good confirmation, but still respect price/stage.
MIXED	Support and contradiction coexist.	Do not assume a clean narrative; investigate important conflicts.
CONFLICT	Multiple meaningful signals disagree with the model side.	Research warning; inspect upstream assumptions and market.
ANOMALY	xK/probability is unusually far from observed context and recent results do not support the side.	Potential real edge OR model/input issue. Requires investigation.
INSUFFICIENT DATA	Not enough reliable context is available.	Avoid overconfidence.
NEWS HOLD	A material lineup/workload/news variable is unresolved.	Wait for resolution before treating the opportunity as actionable.

Why ANOMALY is useful

ANOMALY is a debugging and risk-control label, not a declaration that the model is wrong. It flags unusually large disagreement between the model's current projection and observed recent behavior so that workload, role, lineup, opponent or stale-input assumptions can be checked.

8. Market Movement Labels

Market movement compares the current Trust/Radar snapshot with the previous captured snapshot. It tells you whether the betting market is moving toward or away from the model's preferred side.

Metric	What it means	How to use it
TOWARD MODEL	The market-implied probability moved toward the side the model favors.	Potential confirmation; still ask why it moved.
AGAINST MODEL	The market moved away from the model side.	Investigate news, lineup, workload or sharp disagreement.
STABLE	Very little meaningful implied-probability change (roughly <0.5 percentage points).	No important movement signal.
PRICE MOVED	The side/line is unchanged but the price changed materially.	Recalculate whether EV still clears your threshold.
LINE CHANGED	The strikeout threshold itself changed, e.g. 5.5 to 6.5.	More consequential than a small juice move.
SIDE FLIPPED	Trust's preferred side changed from Over to Under or vice versa.	Always investigate.

BASELINE CAPTURE	No earlier snapshot is available for comparison.	Movement cannot yet be evaluated.
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Do not equate agreement with truth

Market movement toward the model is informative, not proof. Movement against the model is also not proof the model is wrong. The value of this field is contradiction detection.

9. Moneyline: The Core Calculation

The ML model is organized around projected team scoring. It estimates how many runs each team should score using starting-pitcher, bullpen, offense/lineup, park and home-field components. Projected run differential is then transformed into a win probability.

```

Starter run allowance + Bullpen run allowance
x Offense / lineup strength
x Park context
+ Home-field adjustment
= Projected runs for each team

```

```

Projected run differential -> logistic transform -> win probability

```

Projected Runs

Projected Runs is the model's estimated scoring output for each team. The difference between the two teams is the key bridge from baseball inputs to moneyline probability.

Starter Runs

Starter Runs is the portion of projected scoring expected while the opposing starter is in the game. It is informed by the starter's underlying quality profile and the opposing offense. Lower is better for the starter's team.

The starter profile can include measures such as xwOBA, batting average allowed, xSLG, HardHit%, Barrel%, K%, Whiff% and BB% as available in the current model inputs.

Bullpen Runs

Bullpen Runs is the projected scoring after the starter exits. The current bullpen layer uses run prevention, baserunner control, strikeout/walk information and bullpen freshness/rest. This is why two teams with similar starters can still have materially different full-game probabilities.

10. Moneyline: Offense, Workload and Environment

Offense Rating

Offense Rating is centered near 100. A value above 100 indicates an above-average offensive context; below 100 indicates below-average. The lineup model can incorporate wRC+, ISO, xSLG, xwOBA, BB% and K%, while weighting batting-order position and shrinking small hitter samples toward league average.

```

100 = roughly neutral
106 = above-average offensive context
93 = below-average offensive context

```

Official lineups receive greater confidence than active-roster proxies, which is why an ML projection can change meaningfully after lineups are confirmed.

Starter IP

Starter IP is the expected innings from the opposing starting pitcher. In production ML projections this is anchored to the current Pitcher Outs market. It affects both starter exposure and how many innings must be covered by the bullpen.

Park Factor

Park factor scales the expected run environment for both offenses. A value near 1.00 is neutral; above 1.00 is more run-friendly; below 1.00 is more run-suppressing.

```
1.00 = neutral
1.05 = roughly 5% more run-friendly
0.95 = roughly 5% more run-suppressing
```

Home-field Adjustment

The current run model adds approximately +0.20 runs to the home offense before the run differential is converted into win probability.

Run Differential

Run Differential is one of the most useful ML explanation values. It is simply the projected scoring gap between the teams. Larger projected separation leads to a larger win probability, all else equal.

```
DET projected runs: 4.72
COL projected runs: 3.41
Projected differential: +1.31 DET
```

Logistic Scale 1.60

The model converts projected run differential into win probability with a logistic curve. The 1.60 scale controls how aggressively a projected scoring gap translates into probability.

```
P(home win) = 1 / (1 + exp(-RunDiff / 1.60))
```

You generally do not need to handicap the scale manually. The practical interpretation is: projected run difference goes in, win probability comes out.

11. Moneyline Recent Form and Research Context

Team Recent Form

ML Recent Form uses up to the last 10 completed games and considers both winning percentage and run differential per game. Run differential helps distinguish a team that is barely winning from one that is consistently controlling games.

```
Example A: 8-2, +24 run differential -> strongly HOT
Example B: 6-4, -8 run differential -> closer to AVERAGE
```

Separate the components

A single team-form label should never replace the more important game components. A useful ML audit keeps team form, starter form, offense/lineup context and bullpen context distinct so that one hot streak does not hide a weak starting-pitcher or bullpen matchup.

```
TEAM FORM:      HOT
STARTER CONTEXT: AVERAGE
OFFENSE:        HOT
BULLPEN:        COLD
```

12. Fast Reading Order

For a K Prop

1. Compare xK with the sportsbook line.
2. Check Expected IP and Expected BF. Make sure the workload makes sense.
3. Read Structural K. This is the strongest model-side explanation for today.
4. Compare Pitcher K%, Opponent K% and Matchup K%.
5. Look at the Structural / Recent / Season blend.
6. Check Recent Form and Last 10 / Last 5 results.
7. Read workload status and lineup/starter confirmation.
8. Check Market Movement.
9. Read the RCE verdict and confidence.
10. Investigate any CONFLICT, ANOMALY, SIDE FLIPPED or LINE CHANGED flag before betting.

For a Moneyline

11. Compare final model probability with sharp/market probability and available price.
12. Read projected runs and run differential.
13. Inspect starting-pitcher run contribution.
14. Compare offense ratings and lineup status.
15. Inspect bullpen run contribution and availability/freshness context.
16. Check starter workload / expected IP.
17. Confirm whether lineups are official or proxy.
18. Review team Recent Form, but do not let it override the core run model.
19. Check Market Movement and the RCE verdict.
20. Investigate important contradictions before treating the edge as actionable.

13. Worked K Example

Brandon Pfaadt UNDER 4.5 K

MODEL

P(Under): 74.8%
 xK: 3.37
 Structural K: 3.21
 Recent K: 3.60
 Season K: 3.82
 Expected BF: 21.4
 Expected IP: 5.4

RESEARCH

Last 10: 8/10 Under
 Recent Form: COLD
 Opponent profile: favorable to Under
 Workload: NORMAL
 Market: toward model
 Verdict: CONFIRMED

Interpretation: the model is not producing the Under mainly because Pfaadt has recently gone Under. The largest driver is a Structural K around 3.2 based on expected opportunity and matchup. Recent/season production modestly pulls the projection upward to 3.37. The research layer then independently notes that recent outcomes, opponent profile and form broadly support the same direction.

What would make this suspicious?

If xK were 3.2 but Expected IP showed 6.5 innings, Matchup K% was very high, recent production was 7+ Ks per start, and the market was moving hard toward the Over, the RCE should surface a CONFLICT or ANOMALY instead of quietly presenting the Under as clean.

14. Quick Glossary

Metric	What it means	How to use it
xK	Expected strikeouts for today.	Primary K projection.
Structural K	Expected BF x Matchup K%.	Today-specific K foundation.
Recent K	Recency-weighted strikeout production.	Short-term anchor.
Season K	Longer-term strikeout production.	Long-term anchor.
Expected BF	Expected batters faced.	Opportunity volume.
Expected IP	Expected innings pitched.	Workload/leash.
Pitcher K%	Underlying pitcher strikeout skill.	Pitcher side of matchup.
Opponent K%	Expected lineup/team strikeout tendency.	Hitter side of matchup.
Matchup K%	Estimated K probability per batter faced.	Combines pitcher and opponent.
Recent Form	HOT / AVERAGE / COLD relative to own baseline.	Research signal, not re-weighting.
RCE Confidence	Strength/completeness of audit evidence.	Not a win probability.
Projected Runs	Expected team scoring in ML model.	Core ML output.
Starter Runs	Runs expected while opposing starter is in game.	Starter contribution.
Bullpen Runs	Runs expected against bullpen.	Relief contribution.
Offense Rating	Lineup/offense strength centered near 100.	Team hitting context.
Park x	Run-environment multiplier.	Environment adjustment.
Run Differential	Difference between projected team runs.	Input to ML probability transform.
Logistic Scale	Curve parameter converting run gap to win probability.	Math transform, not a handicapping signal.

Bottom line

The purpose of WHY THIS PROBABILITY? is not to make every prediction look explainable after the fact. It is to make the important assumptions inspectable. If a probability looks strange, the screen should now make it possible to determine whether the cause is workload, matchup, pitcher skill, lineup, bullpen, market movement or a potentially bad upstream input.